

Aswin Pradeep C

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Experience

Kireap

April 2025 - July 2026

Backend Engineer | FastAPI, Python, Algorithms | LangGraph, LLMs

Kochi, Kerala

- Cut latency on a CPU-bound dynamic pricing API from 9s to under 2s: profiled hotspots, parallelised computation with ProcessPoolExecutor, batched I/O with asyncio.gather, and added HTTP connection pooling, Redis caching, and exponential backoff with jitter.
- Built agentic pipelines using LangGraph featuring autonomous tool calling, sandboxed code execution, and self correction loops for multi step task completion.

Geojit Financial Services Ltd

Dec 2024 - March 2025

Machine Learning Intern | LLMs, Azure Cognitive Services

Kochi, Kerala

- Developed an LLM-based search system with RAG architecture for enhanced financial data retrieval.
- Automated and optimized manual processes with machine learning to improve efficiency.

Lascade LLC

Jul 2024 - Aug 2024

Backend Developer Intern | Node.js, Express.js, MongoDB, Redis

Kochi, Kerala

- Implemented **RESTful APIs** for the travel module, incorporating input validation and error handling.
- Optimized API performance by implementing **Redis** caching, achieving significant reduction in response times for high-traffic endpoints.

Projects

Picnix | *LangGraph, Gemini, Google Maps, FastAPI* | GitHub

- Built a day trip AI planner on LangGraph orchestrating an 8-node agentic graph that turns a natural language request into a sequenced, weather aware itinerary.
- Designed an LLM first pipeline with intent parsing, place search with relevance gating, route/isochrone building, and dwell time optimization with Arize Phoenix tracing for per run observability and self evaluation.

PitchMailer | *Django, Celery, Redis, OpenAI API, Docker* | GitHub

- Built an AI orchestration platform for automated personalized job outreach. Designed an LLM pipeline for context aware email generation using retrieval-augmented company profiling.
- Architected async workflows with Celery and Redis for multi-stage AI tasks like company research, prompt chaining, and rate limited email delivery. Containerized with Docker and automated deployment workflows.

Additional projects available on **GitHub**

Research Papers

Analyzing the Impact of Modulation Transfer Function (MTF) on Computer Vision: A Multi-Faceted Exploration

- Presented at International Conference on Computations and Data Science(CoDS-2024), IIT Roorkee
- The paper is accepted and set to be published on EquinOCS(Springer Nature).

Technical Skills

Programming Languages: Python, SQL, JavaScript

AI/ML: TensorFlow, LangGraph, RAG

Back-end Tech: Django, FastAPI

Databases: Postgres, Redis

Cloud Services/Platforms: Azure, AWS EC2, Docker, Linux and Self-hosted deployments

Comfortable adopting new stacks as needed. Current focus is on Python/AI backend.

Education

Cochin University of Science and Technology
B.Tech in Computer Science and Engineering

2021 - 2025
First Class with Distinction